

Andrew W. Shenstone

Shenstone, Andrew - Application for Senior Full Stack Engineer @ Topstep

339-223-0797 • shenst1@gmail.com • linkedin.com/in/ashenstone • github.com/shenst1

SUMMARY

Senior full-stack engineer with 12+ years shipping production systems on **TypeScript**, **React**, **Next.js**, **GraphQL**, **REST**, and **PostgreSQL**—the stack this role centers on. I like backbone work: APIs other teams depend on, purchase and subscription flows that stay correct under load, and event-driven patterns (webhooks, background jobs, Inngest) where eventual consistency and race conditions actually matter.

Topstep's payments and purchasing platform maps to work I have done: **Stripe** lifecycle on Plantfolio, payments and KYC-adjacent tooling at Tilia, and commerce integrations at Bloom where reliability and operator clarity were non-negotiable. I bring a production mindset (tests, instrumentation, end-to-end ownership) and I influence patterns through review and pairing, not title inflation.

I have not operated **SQL Server** or prop-firm trading execution at exchange latency; my depth is Postgres, Node/TS services, and money-movement UX. The broader Topstep ecosystem also uses **.NET** in places—I would ramp on house conventions. B.S. Finance (Santa Clara) helps me reason about trader-facing money flows; remote-to-hybrid and on-call participation are alignments I would confirm candidly in process.

SKILLS

Full stack (posting): TypeScript across frontend and backend: React, Next.js, Node.js services, and API design in production, GraphQL and REST in production (Bloom, Tilia, agency work); typed clients and pragmatic schema boundaries, PostgreSQL with Prisma: schema design, migrations, indexing discipline, and MongoDB-to-Postgres migration on Plantfolio, Cloud-native delivery on Vercel/AWS patterns; CI/CD, safe rollouts, and production debugging habits

Payments, async & distributed patterns: Stripe subscriptions and checkout lifecycle in production (Plantfolio); payments/KYC-adjacent tooling at Tilia, Event-style workflows: Inngest, webhooks, background jobs; care for partial failure, idempotency, and reconciliation, High-stakes commerce integrations (Bloom Shopify Plus); operator tooling when money movement must stay correct, SQL Server and exchange-grade low-latency trading stacks are not my headline; transfer is Postgres, async discipline, and finance B.S.

Quality, observability & ownership: Automated testing in production norms (Vitest, Cypress at Plantfolio; strong coverage culture at Tilia), Metrics, logging, and incident-minded ownership from agency scale through founder product delivery, Cross-team collaboration on platform patterns, code review, and raising engineering standards without direct reports

Architecture & async: Monorepos, Async workflows (Inngest, webhooks, background jobs), ETL pipelines, Performance optimization

EDUCATION

B.S. Finance, Santa Clara University — 2010

EXPERIENCE

Founder & Lead Engineer

Plantfolio | November 2023 - Present

AI-native product & engineering

- Built Plantfolio as an AI-native application: plant ID from photos, contextual assistant (Sage), smart tag extraction, and semantic search over 40,000+ species—LLM features customers use in production, not demos
- AI-native way of building: Cursor (top 1% user) and Claude-class agents are my default workflow from early in the product; I own architecture and review while using AI for speed across the 12+ package monorepo

- Designed the monorepo for agent-friendly iteration—TypeScript, clear boundaries, Vitest/Cypress, and consistent patterns so AI-assisted shipping stays maintainable

Product Leadership & Architecture

- Architected and built a comprehensive plant management platform from the ground up, serving as both technical lead and product owner
- Owned the full feature cycle: joined customer calls to identify friction points, architected solutions, implemented, tested, and verified success in production
- Led product strategy and user experience decisions, making product micro-decisions and navigating ambiguity to deliver value iteratively

Technical Achievements

- Modern Frontend Architecture: Implemented Next.js with React, React Server Components, and TypeScript, achieving exceptional performance and developer experience
- Data modeling & migration: Built Plantfolio on MongoDB and shipped production there for 1+ year, then migrated late to PostgreSQL on Neon (15+ interconnected models with Prisma today)—deep familiarity with document and relational modeling tradeoffs
- Database choices: Evaluated Supabase during exploration but did not ship on it; chose Neon for managed Postgres; auth and subscription tiers via Clerk with Postgres-centric access patterns in production today
- Content Management: Created sophisticated content management system with rich text editing, plant tagging, and flexible data structures

Scale & Performance

- Managed a database of 40,000+ plant species with advanced search capabilities using Algolia and vector embeddings
- Implemented comprehensive media management system with AWS S3, supporting image optimization, galleries, and metadata extraction
- Built robust authentication and authorization systems with Clerk, supporting multiple user types and subscription tiers

- Optimized application performance through React Server Components, efficient data fetching, and strategic caching

Client Case Study - AI Signage

- Delivered a production rollout for an independent garden center to automate plant signage generation; one independent retail customer in ongoing production who has been an enthusiastic reference
- Enabled bulk plant import (plain text) to printable signs in seconds; reduced a weekend-long manual process to minutes after initial setup
- Demonstrated ability to translate customer requirements into scalable technical solutions and shorten feedback loops

Full-Stack Ownership

- Developed complete payment integration with Stripe, subscription management, and user onboarding flows
- Built comprehensive email system with React Email, supporting transactional emails, notifications, and marketing campaigns
- Implemented advanced search and filtering capabilities with Algolia, supporting complex plant queries and user preferences

Code Quality & Leadership

- Established coding standards and architectural patterns that enabled rapid, consistent development—with AI-assisted velocity grounded in tests and human review
- Implemented comprehensive testing strategy with Vitest and Cypress, maintaining high code quality
- Built reusable component library and design system, ensuring consistent UI/UX across the platform

Senior Software Engineer

Tilia | January 2023 - August 2023

- Partnered with cross-functional teams in product and design to revamp critical user-facing features
 - Built and shipped work in support of KYC, identity verification, and risk-adjacent tooling as part of Tilia's payments platform (tools team context)
 - Introduced and evolved a GraphQL API on the platform while consuming it from our product—hands-on with both server-side schema/resolvers and client integration
 - Spearheaded and guided a team through the successful adoption of Chakra UI, reducing bespoke components and establishing cohesive design patterns
 - Mentored engineers through pairing and code review in an informal technical lead capacity (no direct reports)
 - Maintained exceptional unit test coverage of over 90% for the Tools team application, alongside comprehensive end-to-end testing
 - Led the swift creation and deployment of a homepage reporting dashboard within the initial two months of joining
 - Role ended in a company-wide layoff affecting approximately one-third of employees
-

Senior Software Engineer

Bloom.wine | September 2020 - October 2022

- Architected a greenfield Next.js application for the primary admin interface of the Bloom product; much of that stack remains in production across many winery clients today
- Engineered Ruby on Rails GraphQL API endpoints shared by multiple applications across mobile and web devices
- CI/CD with CircleCI for build, test, and deployment workflows alongside the Next.js and Rails stack
- Mentored engineers through pairing and code review in an informal technical lead capacity (no direct reports)
- Collaborated with designers using Figma, establishing component library standards usable by both engineering and design

- Integrated DatoCMS and extended application functionality through third-party API integrations (Shopify, Mailchimp, Stripe, Shipstation, ActiveCampaign); all Bloom winery clients were on Shopify Plus
-

Software Engineer II

Tune Inc - Marketplace Automation | August 2019 - May 2020

- Built and operated ETL-style data pipelines and integrations supporting marketplace automation workflows
 - Automated a client workflow using a Node.js/Express API and React.js, reducing the time to complete the task from 10-15 minutes to 10-30 seconds
 - Modernized the primary legacy React.js application by moving from a custom Webpack process to react-scripts, resulting in a 50% reduction in production build size and a 60% reduction in NPM dependencies
 - Reduced development machine CPU usage from 75% to <5% by simplifying the build pipeline and dependency graph
 - Role ended in a company-wide layoff
-

Software Developer

Ply Interactive Inc | June 2014 - July 2019

React Native

- Shipped multiple production React Native applications for agency clients—cross-platform mobile apps delivered end to end
- Washington State Fair: React Native app in sustained multi-year production; very high traffic during fair season, from tens of thousands of users into the hundreds of thousands cumulatively over seasons

Web & platforms

- Developed numerous web applications including Georgetownbeer.com, WashingtonWine.org, TwoWolvesWine.com, Novaerus.com, WellAir, and Plasma-

air.com

- Built a standalone Ruby gem Content Management System, enabling simplified content management within Ruby on Rails projects
- Transitioned the agency to modern technologies like GraphQL, Gatsby.js, and React.js, improving delivery times for new projects
- Bloom.wine Product Development: Built extensive reporting features using complex PostgreSQL queries and interactive filtering, saving clients thousands of hours on previously manual tasks
- Engineered a flexible Wine Club membership structure for wineries, allowing for diverse client configuration and broadening the range of potential sales targets
- Communicated and enforced the standards of Shopify's Polaris React.js component library with designers, resulting in efficiencies gained from leveraging pre-built components

COMPANY REFERENCE

Plantfolio (2023-Present) - AI-native plant management platform (vision, chat, RAG search); built with AI-assisted engineering (Cursor-first) and 40,000+ species catalog - plantfolio.garden/

Tilia (2023) - All-in-one payment platform for digital economies - tilia.io/

Bloom.wine (2020-2022) - Commerce services for the alcohol industry; all clients on Shopify Plus - bloom.wine/

Tune Inc - Marketplace Automation (2019-2020) - SaaS platform for managing marketing partnerships across mobile and web - tune.com/

Ply Interactive Inc (2014-2019) - Boutique agency: web and production React Native apps (e.g. Washington State Fair at major scale); developed Bloom.wine product internally (2017-2019)